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# =====
# MULTIVARIATE ANALYSIS - CREDIT CARD APPROVAL PREDICTION
# =====

# -----
# 1. Import Libraries
# -----
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# -----
# 2. Load Dataset
# -----
file_name = "credit_card.csv"      # Change to your CSV file name
app = pd.read_csv(file_name)

print("="*60)
print("DATASET LOADED SUCCESSFULLY")
print("="*60)

# -----
# 3. Display Basic Information
# -----
print("\nFirst 5 Rows")
print(app.head())

print("\nDataset Shape")
print(app.shape)

print("\nDataset Information")
print(app.info())

print("\nMissing Values")
print(app.isnull().sum())

print("\nStatistical Summary")
print(app.describe())

# -----
# 4. Select Numerical Columns
# -----
numerical_data = app.select_dtypes(include=['int64', 'float64'])

print("\nNumerical Columns")
print(numerical_data.columns.tolist())

# -----
# 5. Correlation Matrix
# -----
correlation_matrix = numerical_data.corr()

print("\nCorrelation Matrix")
print(correlation_matrix)

# -----
# 6. Correlation Heatmap
# -----

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plt.figure(figsize=(14,10))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    linewidths=0.5,
    fmt=".2f"
)

plt.title("Correlation Heatmap")
plt.xticks(rotation=45)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

# -----
# 7. Pair Plot
# -----
print("\nGenerating Pair Plot...")

sns.pairplot(numerical_data)

plt.show()

# -----
# 8. Create Age Column (if DAYS_BIRTH exists)
# -----
if 'DAYS_BIRTH' in app.columns:
    app['AGE'] = abs(app['DAYS_BIRTH']) / 365
    print("\nAGE column created successfully.")

# -----
# 9. Scatter Plot
# Income vs Employment Days
# -----
if 'AMT_INCOME_TOTAL' in app.columns and 'DAYS_EMPLOYED' in app.columns:

    plt.figure(figsize=(8,6))

    sns.scatterplot(
        data=app,
        x='AMT_INCOME_TOTAL',
        y='DAYS_EMPLOYED'
    )

    plt.title("Income vs Employment Days")
    plt.xlabel("Income")
    plt.ylabel("Employment Days")

    plt.show()

# -----
# 10. Scatter Plot
# Age vs Income
# -----
if 'AGE' in app.columns and 'AMT_INCOME_TOTAL' in app.columns:

    plt.figure(figsize=(8,6))

    sns.scatterplot(

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        data=app,
        x='AGE',
        y='AMT_INCOME_TOTAL'
    )

    plt.title("Age vs Income")
    plt.xlabel("Age")
    plt.ylabel("Income")

    plt.show()

# -----
# 11. Boxplot
# Income vs Gender
# -----
if 'CODE_GENDER' in app.columns and 'AMT_INCOME_TOTAL' in app.columns:

    plt.figure(figsize=(8,6))

    sns.boxplot(
        data=app,
        x='CODE_GENDER',
        y='AMT_INCOME_TOTAL'
    )

    plt.title("Income Distribution by Gender")

    plt.show()

# -----
# 12. Boxplot
# Income vs Car Ownership
# -----
if 'FLAG_OWN_CAR' in app.columns and 'AMT_INCOME_TOTAL' in app.columns:

    plt.figure(figsize=(8,6))

    sns.boxplot(
        data=app,
        x='FLAG_OWN_CAR',
        y='AMT_INCOME_TOTAL'
    )

    plt.title("Income by Car Ownership")

    plt.show()

# -----
# 13. Violin Plot
# Property Ownership vs Income
# -----
if 'FLAG_OWN_REALTY' in app.columns and 'AMT_INCOME_TOTAL' in app.columns:

    plt.figure(figsize=(8,6))

    sns.violinplot(
        data=app,
        x='FLAG_OWN_REALTY',
        y='AMT_INCOME_TOTAL'
    )
```

```
plt.title("Income Distribution by Property Ownership")

plt.show()

# -----
# 14. Covariance Matrix
# -----
covariance_matrix = numerical_data.cov()

print("\nCovariance Matrix")
print(covariance_matrix)

# -----
# 15. Correlation with Target Variable
# -----
target_column = None

for col in ['TARGET', 'target', 'Approved', 'APPROVED']:
    if col in app.columns:
        target_column = col
        break

if target_column:

    print(f"\nCorrelation with {target_column}")

    print(
        app.corr(numeric_only=True)[target_column]
        .sort_values(ascending=False)
    )

# -----
# 16. Save Correlation Matrix
# -----
correlation_matrix.to_csv("Correlation_Matrix.csv")

print("\nCorrelation matrix saved as 'Correlation_Matrix.csv'")

# -----
# 17. End of Analysis
# -----
print("\n" + "="*60)
print("MULTIVARIATE ANALYSIS COMPLETED SUCCESSFULLY")
print("="*60)
```

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-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_4684/501192741.py in <cell line: 0>()
    15 # -----
    16 file_name = "credit_card.csv"          # Change to your CSV file name
--> 17 app = pd.read_csv(file_name)
    18
    19 print("="*60)

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4 frames

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/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode,
encoding, compression, memory_map, is_text, errors, storage_options)
    871         if ioargs.encoding and "b" not in ioargs.mode:
    872             # Encoding
--> 873             handle = open(
    874                 handle,
    875                 ioargs.mode,

```

FileNotFoundError: [Errno 2] No such file or directory: 'credit_card.csv'

Next steps: [Explain error](#)

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# =====
# MULTIVARIATE ANALYSIS (WITHOUT LOADING OR UPLOADING CSV)
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# -----
# Check if dataset exists
# -----
try:
    app
except NameError:
    raise Exception("Dataset 'app' is not found. Please create or load the DataFrame first.")

print("="*60)
print("MULTIVARIATE ANALYSIS")
print("="*60)

# -----
# Basic Information
# -----
print("\nFirst 5 Rows")
print(app.head())

print("\nDataset Shape")
print(app.shape)

print("\nColumn Names")
print(app.columns.tolist())

print("\nData Types")
print(app.dtypes)

print("\nMissing Values")
print(app.isnull().sum())

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```
print("\nStatistical Summary")
print(app.describe(include='all'))

# -----
# Select Numerical Columns
# -----
numerical_data = app.select_dtypes(include=['int64', 'float64'])

print("\nNumerical Columns")
print(numerical_data.columns.tolist())

# -----
# Correlation Matrix
# -----
correlation_matrix = numerical_data.corr()

print("\nCorrelation Matrix")
print(correlation_matrix)

# -----
# Heatmap
# -----
plt.figure(figsize=(12,8))
sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)
plt.title("Correlation Heatmap")
plt.show()

# -----
# Pair Plot
# -----
if len(numerical_data.columns) > 1:
    sns.pairplot(numerical_data)
    plt.show()

# -----
# Covariance Matrix
# -----
covariance_matrix = numerical_data.cov()

print("\nCovariance Matrix")
print(covariance_matrix)

# -----
# Scatter Plot (First Two Numerical Columns)
# -----
if len(numerical_data.columns) >= 2:
    plt.figure(figsize=(8,6))
    sns.scatterplot(
        x=numerical_data.columns[0],
        y=numerical_data.columns[1],
        data=app
    )
    plt.title(f"{numerical_data.columns[0]} vs {numerical_data.columns[1]}")
    plt.show()
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# -----
# Boxplot (First Numerical Column)
# -----
plt.figure(figsize=(8,6))
sns.boxplot(y=numerical_data.iloc[:,0])
plt.title(f"Boxplot of {numerical_data.columns[0]}")
plt.show()

# -----
# Histogram (First Numerical Column)
# -----
plt.figure(figsize=(8,6))
sns.histplot(numerical_data.iloc[:,0], bins=20, kde=True)
plt.title(f"Histogram of {numerical_data.columns[0]}")
plt.show()

# -----
# Save Correlation Matrix
# -----
correlation_matrix.to_csv("Correlation_Matrix.csv")

print("\nCorrelation Matrix saved successfully.")
print("\nMULTIVARIATE ANALYSIS COMPLETED SUCCESSFULLY.")

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NameError Traceback (most recent call last)
[/tmp/ipykernel_4684/1810676979.py](#) in <cell line: 0>()

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13 try:
---> 14     app
15 except NameError:

```

NameError: name 'app' is not defined

During handling of the above exception, another exception occurred:

Exception Traceback (most recent call last)
[/tmp/ipykernel_4684/1810676979.py](#) in <cell line: 0>()

```

14     app
15 except NameError:
---> 16     raise Exception("Dataset 'app' is not found. Please create or load the DataFrame
first.")
17
18 print("="*60)

```

Exception: Dataset 'app' is not found. Please create or load the DataFrame first.

Next steps: [Explain error](#)

```

# =====
# MULTIVARIATE ANALYSIS (NO CSV FILE REQUIRED)
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# -----
# Create Sample Credit Card Dataset
# -----
app = pd.DataFrame({

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'AGE': [25, 30, 35, 40, 45, 50, 28, 32, 38, 42],
'AMT_INCOME_TOTAL': [25000, 35000, 45000, 55000, 65000,
                     75000, 30000, 40000, 50000, 60000],
'DAYS_EMPLOYED': [500, 1000, 1500, 2000, 2500,
                  3000, 1200, 1800, 2200, 2800],
'CNT_FAM_MEMBERS': [2, 3, 4, 5, 3, 2, 4, 3, 5, 2],
'CNT_CHILDREN': [0, 1, 2, 1, 3, 2, 0, 1, 2, 1],
'TARGET': [1, 0, 1, 1, 0, 1, 0, 1, 0, 1]
})

print("Dataset Loaded Successfully\n")

# -----
# First Five Records
# -----
print(app.head())

# -----
# Dataset Information
# -----
print("\nShape of Dataset:", app.shape)

print("\nColumn Names:")
print(app.columns)

print("\nData Types:")
print(app.dtypes)

print("\nMissing Values:")
print(app.isnull().sum())

print("\nStatistical Summary:")
print(app.describe())

# -----
# Correlation Matrix
# -----
correlation = app.corr()

print("\nCorrelation Matrix:")
print(correlation)

# -----
# Heatmap
# -----
plt.figure(figsize=(8,6))
sns.heatmap(correlation,
            annot=True,
            cmap='coolwarm',
            fmt=".2f")

plt.title("Correlation Heatmap")
plt.show()

# -----
# Pair Plot
# -----
sns.pairplot(app)
plt.show()

# -----

```



```
# Scatter Plot
# -----
plt.figure(figsize=(7,5))
sns.scatterplot(data=app,
                x='AGE',
                y='AMT_INCOME_TOTAL',
                hue='TARGET')

plt.title("Age vs Income")
plt.show()

# -----
# Box Plot
# -----
plt.figure(figsize=(6,5))
sns.boxplot(data=app,
            x='TARGET',
            y='AMT_INCOME_TOTAL')

plt.title("Income vs Target")
plt.show()

# -----
# Covariance Matrix
# -----
covariance = app.cov()

print("\nCovariance Matrix:")
print(covariance)

print("\nMultivariate Analysis Completed Successfully")
```

